

IEOR 4101: Probability, Statistics and Simulation

Lecture 8: Special Distributions and Random Variables

Discrete Uniform Random Variable

- A discrete uniform random variable X on $\{a, a+1, \dots, b\}$ is equally likely to be any x in $\{a, a+1, \dots, b\}$

$\{1, 2, \dots, 6\}$

- E.g., fair die roll; fair coin toss

$\{0, 1\}$

- $P(X = x) = 1/(b-a+1), x = a, a+1, \dots, b$

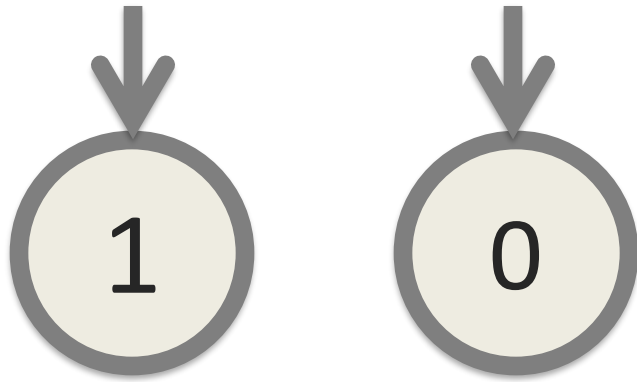
- $E[X]; \text{Var}(X)$

$$\mu = E[X] = \frac{1}{b-a+1} (a + (a+1) + \dots + b)$$

$$\text{Var}(X) = \frac{1}{b-a+1} ((a-\mu)^2 + \dots + (b-\mu)^2)$$

Bernoulli Random Variables

- A Bernoulli random variable is used to indicate the success/failure of an event, with a success probability p ; e.g., a coin toss
- Sometimes called Bernoulli trial
- $P(X = 1) = p$; $P(X = 0) = 1-p$

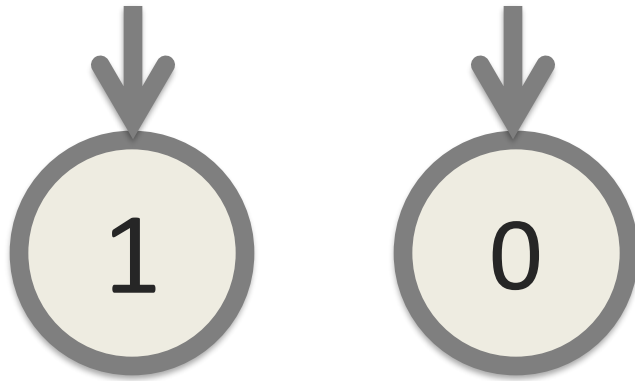


$$E[X] = 1 \times P(X=1) + 0 \times P(X=0) = P(X=1) = p$$

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 = E[X] - (E[X])^2 \\ &= p - p^2 = p(1-p) \end{aligned}$$

Bernoulli Random Variables

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- $P(X = 1) = p$; $P(X = 0) = 1-p$



- $E[X] = 1 \times P(X=1) + 0 \times P(X=0) = p$
- $\text{Var}(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1 - p)$
- Notation: $\text{Ber}(p)$

Binomial Distribution

- Consider n independent repetitions of a Bernoulli trial with success probability p
- X denotes the number of successes in n trials
- Pmf: $P(X = k) = p(k) = \binom{n}{k} p^k (1-p)^{n-k}$, $k=0, \dots, n$

position of trials



pick k positions out of n positions.

This gives $\binom{n}{k}$ possibilities.

Express $X = X_1 + \dots + X_n$ where
 $X_i \sim \text{Ber}(p)$, independent.

$$E[X] = E[X_1] + \dots + E[X_n] = np$$

$$\text{var}(X) = \text{var}(X_1 + \dots + X_n)$$

$$= \text{var}(X_1) + \dots + \text{var}(X_n)$$

$$= np(1-p)$$

because X_i 's
are independent

Binomial Distribution

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- $E[X] = np$
- $\text{Var}(X) = np(1-p)$
- Denoted by $\text{Bin}(n, p)$

Binomial Distribution

- **Example:** Disks produced by a certain company will be defective with probability 0.01 independently of each other. The company sells disks in packages of 10 and offers a money-back guarantee that at most 1 out of 10 disks is defective. What proportion of packages is returned? What is the proportion of packages having exactly two defective disks?

$$X = \# \text{ defective}$$

$$\sim \text{Bin}(10, 0.01)$$

$$\begin{aligned} P(\text{return}) &= P(X > 1) = 1 - P(X = 0) - P(X = 1) \\ &= 1 - (1 - 0.01)^{10} - 10 \times 0.01 \times (1 - 0.01)^{10-1} \end{aligned}$$

$$P(X=2) = \binom{10}{2} 0.01^2 (1-0.01)^{10-2}$$

$$= \frac{10 \times 9}{2} 0.01^2 0.99^8$$

Geometric Distribution

- $X = \#$ of Bernoulli trials with success probability p until we see a first success

Geometric Distribution

- $X = \#$ of Bernoulli trials with success probability p until we see a first success
- Pmf: $P(X = k) = (1-p)^{k-1}p$, $k = 1, 2, \dots$
- $E[X] = 1/p$
- $\text{Var}(X) = (1-p)/p^2$
- Denoted by $\text{Geo}(p)$

Geometric Distribution

- **Example:** One die is rolled over and over. How many rolls are required so that the probability of rolling a six within this number is at least 50%?

Find k such that

$$P(X \leq k) \geq 50\%$$

$X = \#$ rolls needed to see "6"

$$X \sim \text{Geo}\left(\frac{1}{6}\right)$$

$$\begin{aligned} P(X \leq k) &= 1 - P(X > k) \\ &= 1 - (1-p)^k \end{aligned}$$

because $X > k$ is
the same as
no success in the
first k trials

Find k such that

$$P(X \leq k) \geq 5\%$$

or

$$1 - (1-p)^k \geq 5\%$$

$$5\% \geq (1-p)^k$$

$$\log \frac{1}{2} \geq k \log(1-p)$$

$$k \geq \frac{\log \frac{1}{2}}{\log(1-p)}$$

Geometric Distribution

- **Example:** One die is rolled over and over. How many rolls are required so that the probability of rolling a six within this number is at least 50%?
- **Solution.** Let this number be k . Let X be a $\text{Geo}(1/6)$ random variable. We are interested in the value of k so that $P(X \leq k) \geq 0.5$
- Or $P(X > k) \leq 0.5$
- $P(X > k) = (1-p)^k = (5/6)^k \leq 0.5 \rightarrow k \geq 3.8$
- So smallest k is 4

Geometric Distribution

- **Example.** Tom and his mom lost each other while wandering through a crowded amusement park. Their iphones are out of battery. Tom's mom, being a wise woman, foresaw this and made a prior agreement with Tom. Tom will wait at the entrance of one of the 15 attractions and his mom searches the attractions in a random order (the “wait-for-mommy” strategy). What are the expected value and standard deviation of the number of searches before Tom's mom finds him?

Geometric Distribution

- **Example.** Tom and his mom lost each other while wandering through a crowded amusement park. Their iphones are out of battery. Tom's mom, being a wise woman, foresaw this and made a prior agreement with Tom. Tom will wait at the entrance of one of the 15 attractions and his mom searches the attractions in a random order (the “wait-for-mommy” strategy). What are the expected value and standard deviation of the number of searches before Tom's mom finds him?
- Suppose now that Tom and his mom agrees that, in intervals of 10 minutes, they will each pick a random entrance and go to their chosen entrances. What are the expected value and standard deviation of the number of searches in this case?

Poisson Distribution

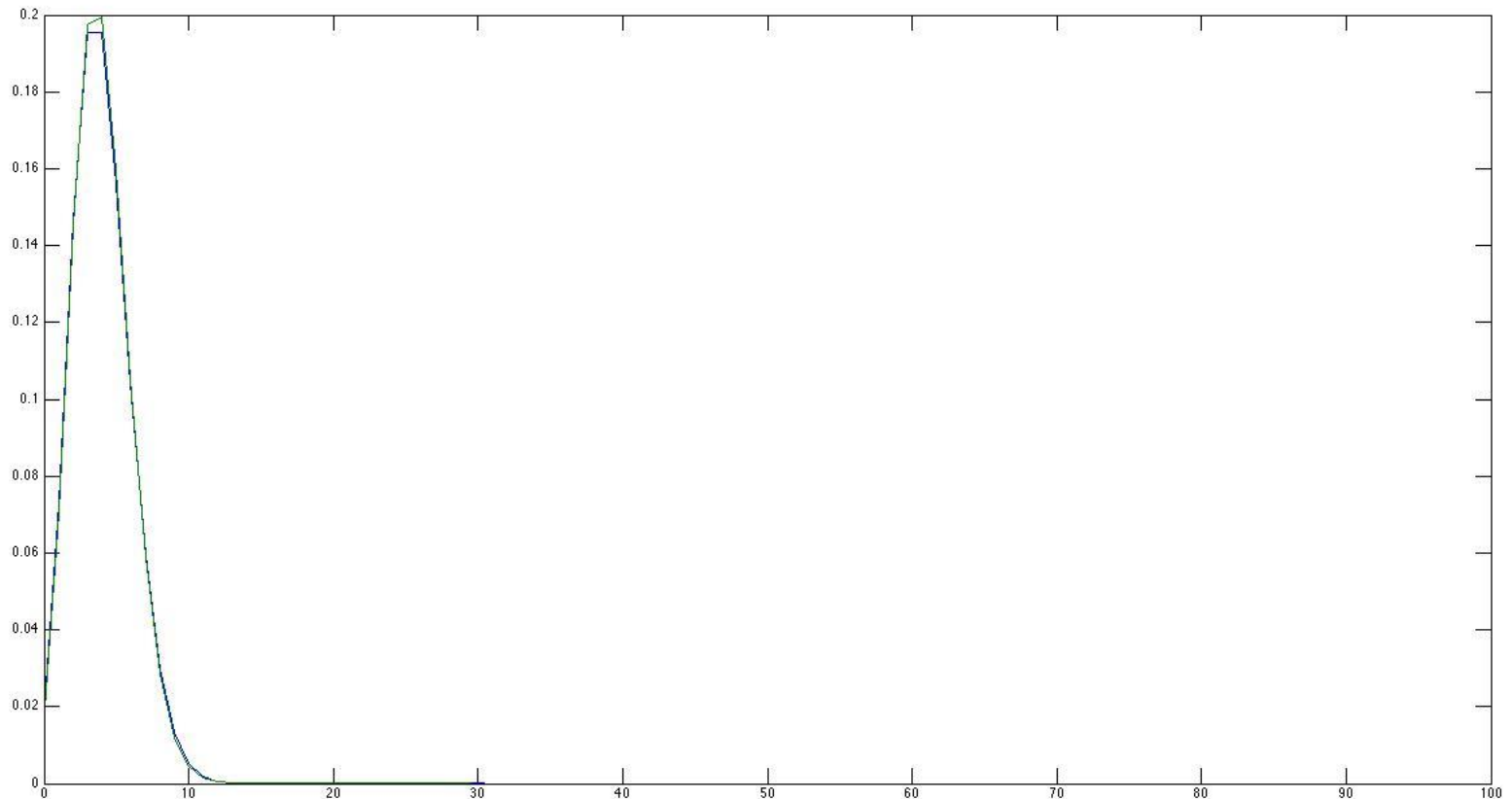
- Poisson r.v. counts the number of occurrences of interest in a segment of time or space; possible values are 0, 1, 2, ...
- Poisson distribution with parameter $\lambda = np$ approximates $\text{Bin}(n, p)$ when n is large and p is small; e.g., number of calls received by an emergency center
- X has a Poisson distribution with parameter $\lambda > 0$ if its pmf satisfies

$$P(X = k) = e^{-\lambda} \lambda^k / k!, \quad k = 0, 1, 2, \dots$$

- $E[X] = \text{Var}(X) = \lambda$
- Denoted by $\text{Poi}(\lambda)$

Poisson Distribution

- Plot shows $\text{Poi}(\lambda)$ with $\lambda = 4$ and $\text{Bin}(n, p)$ with $n = 100$, $p = 0.04$
- Blue is Poisson pmf and green is Binomial pmf



Poisson Distribution

- **Example.** At Whole Foods Grocery, customers are thought to arrive at the checkout section at a rate of **16 customers per hour**. The store manager believes that the service time for each customer is **constant at six minutes**. Suppose that during a **six minute interval, three checkers are available**, i.e., three customers can be served during a six-minute time segment. The manager is interested in the probability that one or more customers will have to wait for service during a six-minute window. Determine this probability, supposing that the number of arrivals in all time segment is a Poisson random variable.

Hypergeometric Distribution

- Hypergeometric distributions are often used in situations where we are sampling *without* replacement in a finite population
- Population has two mutually exclusive attributes
 - E.g., male vs female, positive vs negative to a blood test, etc
- Want to count the number in the sample with a certain attribute
 - E.g., number of male, number of positive reactions to a blood test, etc

Hypergeometric Distribution

- For a population of size N , suppose that there are K “successes”
 - $N = 30$ and $K = 12$ in previous example
- A random sample size of n
- X = number of successes in the sample
- $P(X = k) = \left[\binom{K}{k} \times \binom{N-K}{n-k} \right] / \binom{N}{n}$

of ways to pick
the k successes

of ways to pick
the $n-k$ failures

of ways to pick
the sample

Hypergeometric Distribution

- **Example (Gender equity).** Suppose a company needs to downsize one department having 30 people – 12 women and 18 men. 10 people were laid off, and upper management said that the layoffs were done randomly. Of the 10 people, 8 were women. A labor attorney is interested in the probability of 8 or more women being laid off by chance

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- **Solution.** The probability of having more than 8 women by chance is

$$\frac{\binom{12}{8} \times \binom{18}{2}}{\binom{30}{10}} + \frac{\binom{12}{9} \times \binom{18}{1}}{\binom{30}{10}} + \frac{\binom{12}{10}}{\binom{30}{10}} = 0.0026$$

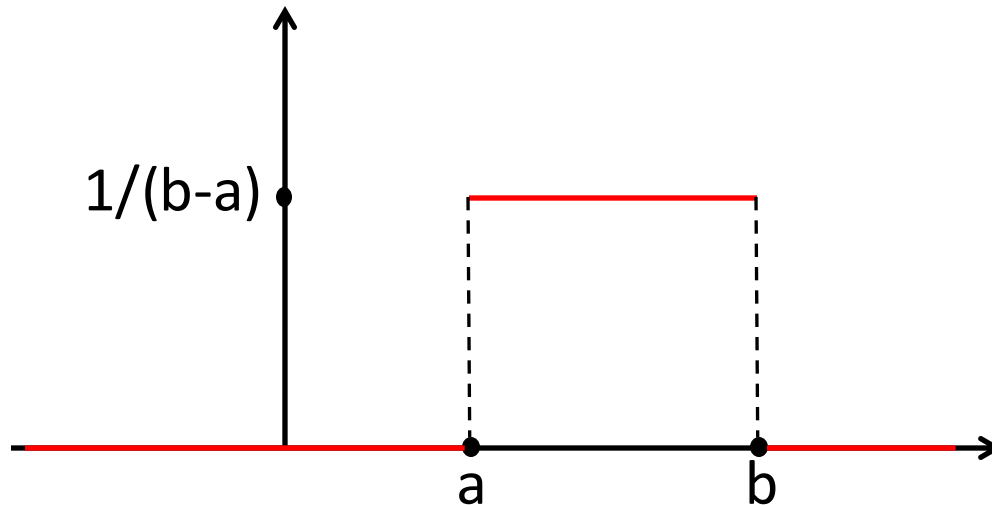
Hypergeometric Distribution

- When the population size N is large compared to the sample size n , we can approximate the hypergeometric distribution using a binomial distribution.
- In previous example, if the department has say 300 people, of which 120 are women and 180 are men. What is the probability of having 8 women in a random sample of size 10?
- Binomial approximation: probability of women = $120/300 = 0.4$
- $P(8 \text{ women}) \approx \binom{10}{8} \times 0.4^8 \times 0.6^2 = 0.01$
- Calculation using hypergeometric distribution gives 0.0097

Uniform Random Variables

- A random variable X is uniformly distributed over $[a, b]$ if its probability density function (pdf) $f(x)$ is

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$



- Denote it by $\text{Unif}(a, b)$
- Used to model a “completely random” selection over the interval

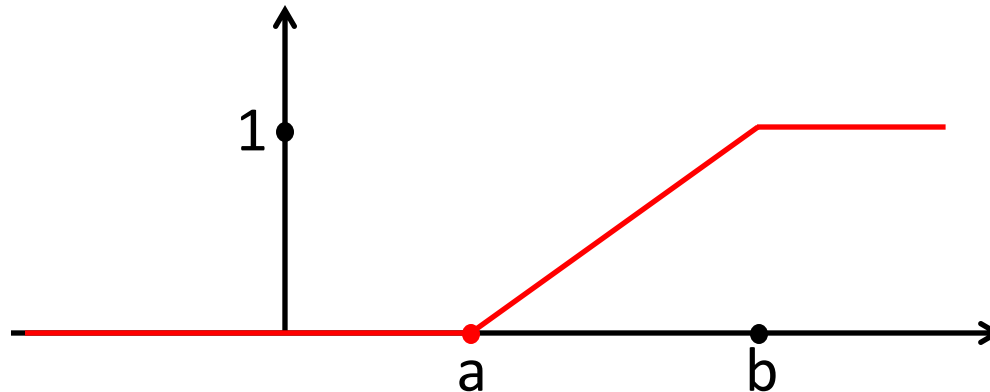
Uniform Random Variables

- Note that this is a valid pdf because

$$\int_a^b \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b dx = \frac{1}{b-a} \cdot (b-a) = 1$$

- The (cumulative) distribution function of $\text{Unif}(a, b)$ is

$$F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 1 & \text{if } x > b. \end{cases}$$



Uniform Random Variables

- Expectation of $X \sim \text{Unif}(a, b)$ is

$$\mathbb{E}[X] = \int_a^b x \cdot \frac{1}{b-a} dx = \left. \frac{x^2}{2(b-a)} \right|_a^b = \frac{a+b}{2}$$

- Variance of X can be computed as follows

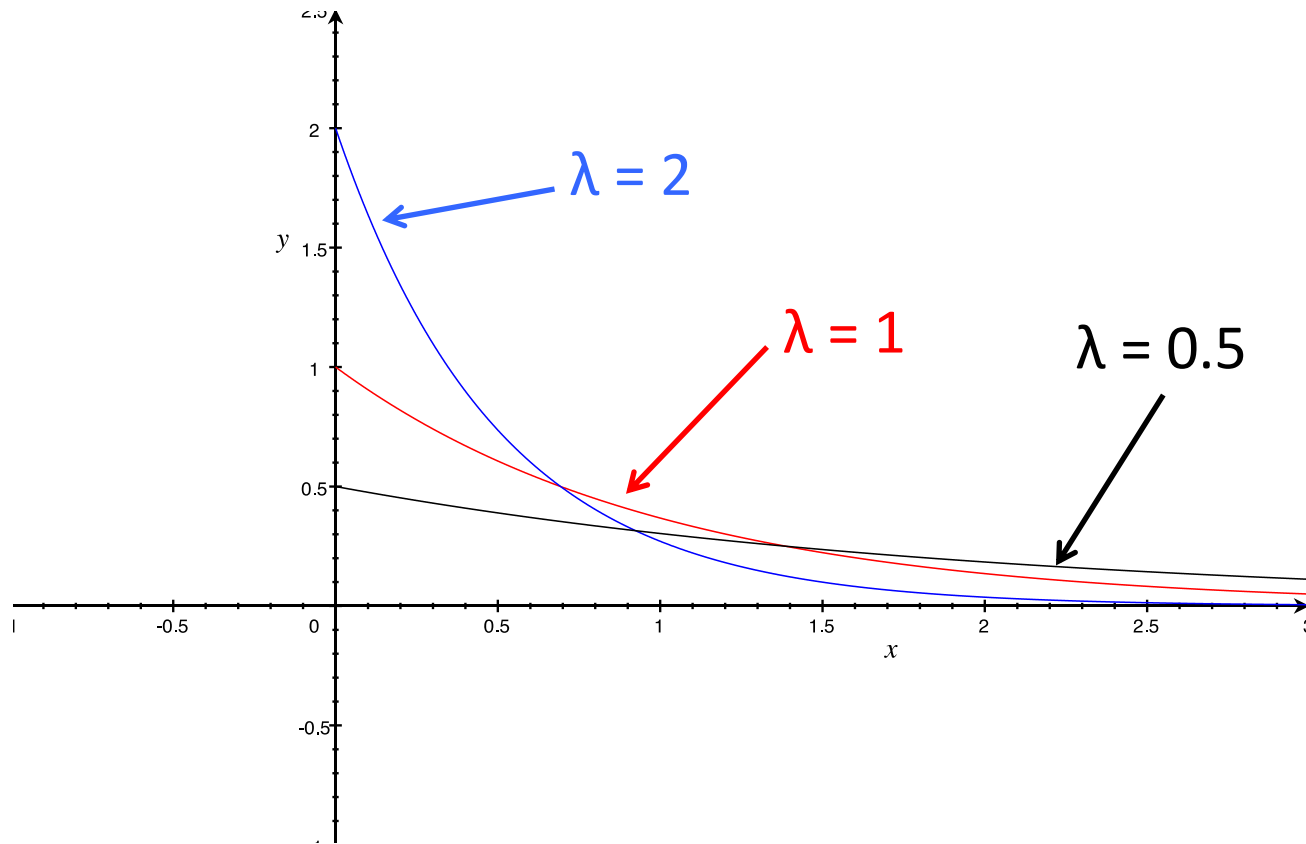
$$\mathbb{E}[X^2] = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \left. \frac{x^3}{3(b-a)} \right|_a^b = \frac{b^3 - a^3}{3(b-a)}$$

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{b^3 - a^3}{3(b-a)} - \left(\frac{a+b}{2} \right)^2 \\ &= \frac{b^2 + ab + a^2}{3} - \frac{a^2 + 2ab + b^2}{4} = \frac{(b-a)^2}{12} \end{aligned}$$

Exponential Random Variables

- A random variable X has an *exponential* pdf with parameter $\lambda > 0$ if the pdf $f(x)$ is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

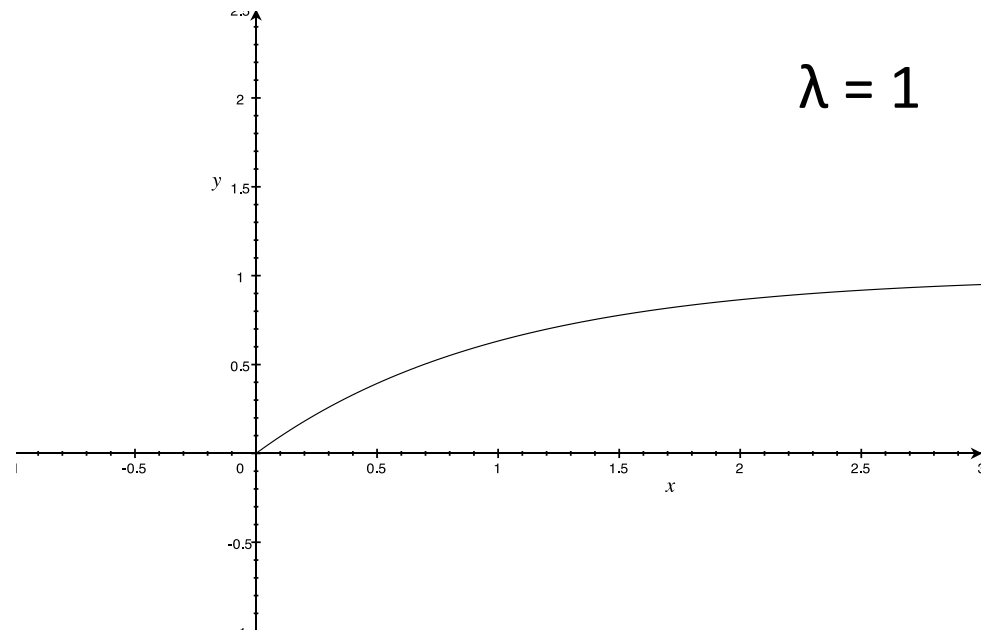


Exponential Random Variables

- Denoted by $\text{Exp}(\lambda)$
- Often used to model the time till some event (e.g., earthquake, phone call, radioactive decay, etc) occurs
- cdf $F(x)$ is given by

$$F(x) = \int_0^x \lambda e^{-\lambda y} dy = -e^{-\lambda y} \Big|_0^x = 1 - e^{-\lambda x}, x \geq 0$$

- $E[X] = 1/\lambda$
- $\text{Var}(X) = 1/\lambda^2$



Exponential Random Variables

- **Memoryless property.** Let X be an exponential rv. Then for any positive s and t , $P(X - s > t \mid X > s) = P(X > t)$
- In words: think of X as the time till some event occurs. I have observed for s amount of time and don't see the occurrence of the event. Conditioned on this, the random amount of time that I have to wait *further* (i.e., $X-s$) till the occurrence has the same distribution as X

Exponential Random Variables

- Proof of memoryless property:

$$\begin{aligned}\mathbb{P}(X - s > t \mid X > s) &= \frac{\mathbb{P}(X > s + t, X > s)}{\mathbb{P}(X > s)} \\ &= \frac{\mathbb{P}(X > s + t)}{\mathbb{P}(X > s)} \\ &= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t}\end{aligned}$$

Exponential Random Variables

- **Property 2:** Suppose $X_1 \sim \text{Exp}(\lambda_1)$, $X_2 \sim \text{Exp}(\lambda_2)$, ..., $X_n \sim \text{Exp}(\lambda_n)$ are independent. Let $Y = \min\{X_1, X_2, \dots, X_n\}$. Then $Y \sim \text{Exp}(\lambda_1 + \lambda_2 + \dots + \lambda_n)$.

- Proof. For any $s > 0$,

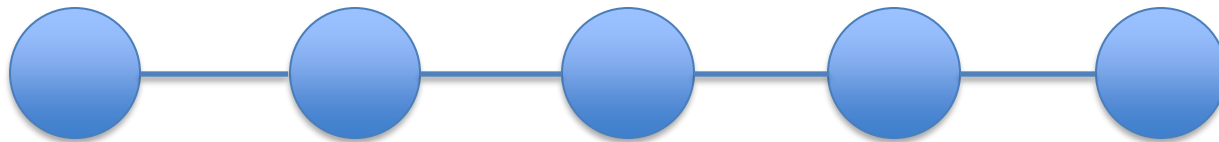
$$\begin{aligned}\mathbb{P}(Y > s) &= \mathbb{P}(\min\{X_1, X_2, \dots, X_n\} > s) \\ &= \mathbb{P}(X_1 > s, X_2 > s, \dots, X_n > s) \\ &= \mathbb{P}(X_1 > s)\mathbb{P}(X_2 > s) \cdots \mathbb{P}(X_n > s) \\ &= e^{-\lambda_1 s} \cdots e^{-\lambda_n s} \\ &= e^{-(\lambda_1 + \cdots + \lambda_n)s}\end{aligned}$$

Exponential Random Variables

- **Example.** A crew of workers have 3 interchangeable machines A, B and C. They need 2 of them to be functional simultaneously to do their job. The lifetime of the machines are independently distributed as $\text{Exp}(\lambda)$. They start A and B first and keep C as reserve to replace whichever of A or B that breaks down first. They then continue until only one of the machines remains functional, at which time they are forced to stop. What is the expected time to stop? What is the probability that at this time, the only functional machine is C?

Exponential Random Variables

- **Example.** A series system has 5 components. The system is functional if and only if *all* of its components are functional. Suppose that the lifetimes of the components are independent exponential random variables with mean 10 minutes each. What is the probability that the system survives for 2 minutes?



Exponential Random Variables

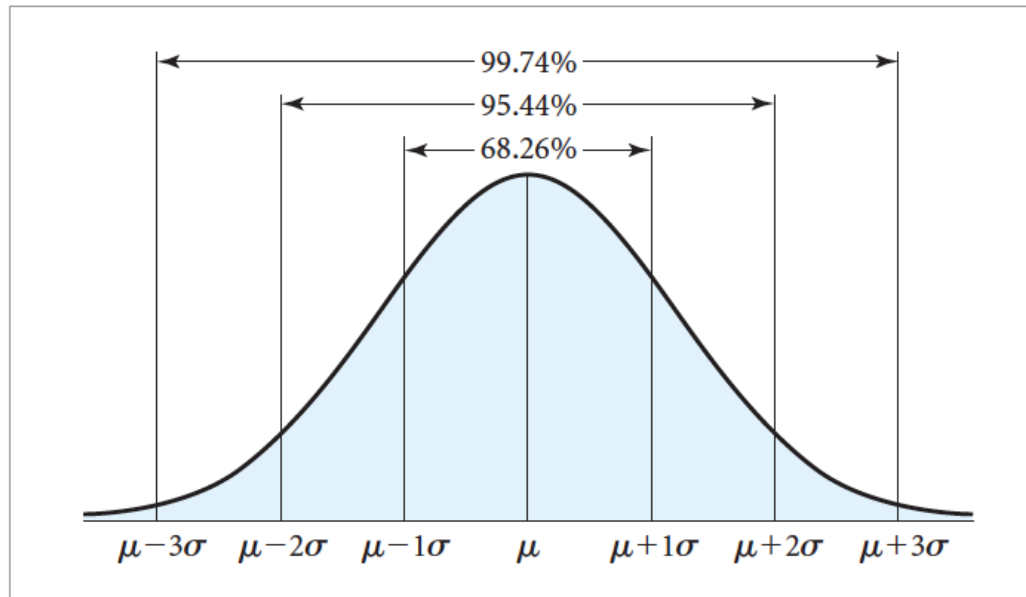
- **Example.** A series system has 5 components. The system is functional if and only if *all* of its components are functional. Suppose that the lifetimes of the components are independent exponential random variables with mean 10 minutes each. What is the probability that the system survives for 2 minutes?
- **Solution.** Let X_i be the lifetime of the i^{th} component, $i=1, 2, \dots, 5$. Then, $X_1, X_2, \dots, X_5$ are independent exponential random variables with parameter $1/10$ each. The failure time of the system is $Y = \min\{X_1, X_2, \dots, X_5\}$, an exponential random variable with parameter $5 \times 1/10 = 1/2$ by **Property 2**. Thus, the desired probability is

$$\mathbb{P}(Y \geq 2) = \int_2^{\infty} \frac{1}{2} e^{-\frac{x}{2}} dx = -e^{-\frac{x}{2}} \Big|_{x=2}^{\infty} = e^{-1}$$

Normal Random Variables

- A bell-shaped pdf curve
- Random variable X is normally distributed with mean μ and variance σ^2 if it has pdf

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, x \in \mathbb{R}$$



Normal Random Variables

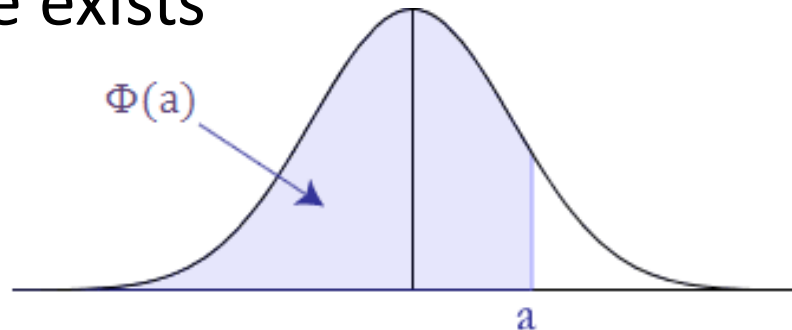
- Normal distribution and r.v. show up in many natural phenomena and are important in statistics (later)
- $E[X] = \mu$, $\text{Var}(X) = \sigma^2$
- Denote it by $N(\mu, \sigma^2)$
- $N(0, 1)$ is called the standard normal distribution

Normal Random Variables

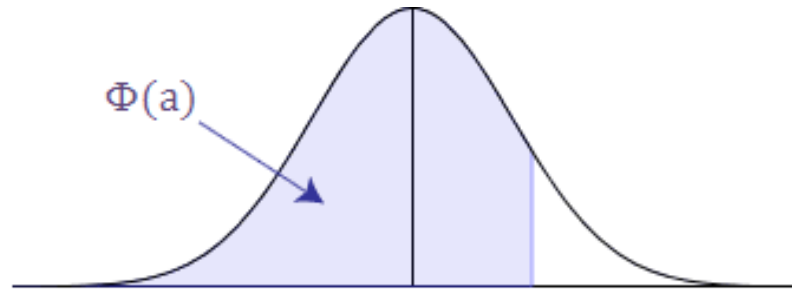
- **Property 1:** If $X \sim N(\mu, \sigma^2)$, then $aX+b \sim N(a\mu+b, a^2\sigma^2)$
- **Property 1':** If $X \sim N(\mu, \sigma^2)$, then $Z = (X-\mu)/\sigma \sim N(0, 1)$
- Moral: all normal rv can be “reduced” to the standard normal
- Often use $\Phi(x)$ to denote the cdf of standard normal

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy$$

- No closed-form formula for $\Phi(x)$ but the standard normal table exists



Normal Random Variables



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9031	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817

Normal Random Variables

- **Example.** After completing a study, the Chicago O'Hare Airport managers have concluded that the time needed to get passengers loaded onto an airplane is (approximately) normally distributed with mean 15 minutes and standard deviation 3.5 minutes. What is the probability that a flight will take at least 22 minutes to get a passenger loaded?

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- **Solution.** Let X be the random time. Then $X \sim N(15, 3.5^2)$. So, $(X-15)/3.5 \sim N(0, 1)$.
- $P(X \geq 22) = P\left(\frac{X-15}{3.5} \geq \frac{22-15}{3.5}\right) = P(Z \geq 2) = 1 - \Phi(2)$
- $\Phi(2) = 0.9772$; so $P(X \geq 22) = 1 - 0.9772 = 0.0228$

Normal Random Variables

- **Property 2:** if $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent normal rv, then $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

Normal Random Variables

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- **Example.** Data from the National Oceanic and Atmospheric Administration indicate that the yearly precipitation in Los Angeles is a normal rv with mean 12.08 inches and standard deviation 3.1 inches. What is the probability that the total precipitation during the next 2 years will exceed 30 inches? Assume precipitations are independent across years.

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- **Solution.** Let X_1 and X_2 be the precipitations for the next 2 years. $X_1, X_2 \sim N(12.08, 3.1^2)$ are independent. Then $X_1 + X_2 \sim N(2 \times 12.08, 2 \times 3.1^2)$.
- $$P(X_1 + X_2 > 30) = P\left(\frac{X_1 + X_2 - 2 \times 12.08}{\sqrt{2 \times 3.1^2}} > \frac{30 - 2 \times 12.08}{\sqrt{2 \times 3.1^2}}\right) = P(Z > 1.33) = 1 - \Phi(1.33) = 1 - 0.9082 = 0.0918$$